

AIQT CLI

Milestone 6 Build Specification v0.4

Review and Export Engine

Build-planning document for implementing AIQT Milestone 6. Not a runtime document model for AIQT user projects.

Field	Value
Product	AIQT CLI
Specification	Milestone 6 Build Specification
Version	v0.4
Milestone	M6 - Review and Export Engine
Aligns With	AIQT MVP Specification v0.6 and Technical Architecture Specification v0.3
Depends On	M1 - Canonical State Foundation, tag m1-canonical-state-foundation M2 - Project Context Capture, tag m2-project-context-capture M3 - Work Graph Planning Foundation, tag m3-work-graph-planning M4 - Agent Handoff Packet Engine, tag m4-agent-handoff-packet M5 - Checkpoint Engine, merge commit 41a8e49, tag m5-checkpoint-engine
Primary Agent Target	Claude Code or equivalent coding agent
Implementation Status	Ready for final review before build handoff
Document status	v0.4 build-handoff candidate. Clarifies dependency-reference review findings, removes project-summary from M6 targets, fixes finding category naming, and adds multi-file export runlog examples.

Document status: Ready for final review before build handoff. v0.4 resolves M6 v0.3 review findings and keeps review/export behavior deterministic.

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1. Source Alignment

This document is derived from AIQT MVP Specification v0.6, AIQT Technical Architecture Specification v0.3, and the completed milestone sequence M1 through M5. It assumes M5 is merged to main at commit 41a8e49 and tagged as m5-checkpoint-engine. The repository is expected to contain the checkpoint engine, state.checkpoints records, current-work-cycle transitions, dependency readiness recalculation, and all tests from M1 through M5.

Milestone tags are literal repository tags, not title-derived names. Use these actual tags: m1-canonical-state-foundation, m2-project-context-capture, m3-work-graph-planning, m4-agent-handoff-packet, and m5-checkpoint-engine.

The governing design remains:

```
state -> graph -> packet -> checkpoint -> review/export
not:
spec -> milestone file -> task file -> review file -> another markdown file
```

M6 implements the review and export step. Review inspects canonical state and reports findings. Export generates optional human-readable markdown views under .aiqt/exports/. M6 must not create markdown as a source of truth, must not mutate workflow state during review, and must not implement replanning or automated repair.

1.1 v0.4 Change Summary

- Keeps the v0.2/v0.3 fix that normal workflow positions are non-blocking review findings, not review failures.
- Scopes work-unit quality blockers to actionable statuses: ready, in_progress, needs_review, and done.
- Makes planned-work-unit quality gaps non-blocking medium findings so review remains CI-safe during normal planning.
- Restores explicit export nextRecommendedCommand rules for dry-run, write success, invalid input, soft skips, and current workflow position.
- Restores export target content minimums for project-plan, technical-spec, status-report, and agent-packet exports.
- Defines recommendedExportTargets computation using only valid M6 export targets and adds it to the example review JSON.
- Classifies broken dependency references discovered by review as critical blocking integrity findings with exit code 1.
- Uses category = quality consistently for work-unit quality findings.
- Adds explicit export.generated runlog example for aiqt export all writing multiple files.
- Adds regression tests for planned-work-unit non-blocking review behavior, ready-work-unit blocking handoff-quality behavior, and broken dependency reference handling.
- Keeps export as generated documentation only; canonical state remains project.json and state.json.

2. Milestone Objective

Milestone 6 implements aigt review and aigt export. It turns canonical workflow state into actionable quality feedback and optional human-readable documents without changing the canonical source of truth.

Milestone 6 must prove that AIQT can:

- run aigt review against any initialized AIQT project;
- inspect project context, work graph, work units, dependencies, checkpoints, current pointers, packet metadata, and runlog health;
- report normal workflow position without treating it as review failure;
- detect invalid state, broken graph references, stale execution cycles, missing checkpoint coverage, unresolved review work, and export readiness gaps;
- return deterministic review findings through human-readable output and CommandResult JSON;
- run aigt export for supported markdown views;
- write optional generated files only under .aiqt/exports/;
- support --dry-run for export without file mutation or runlog events;
- append export.generated runlog events only when export writes files;
- preserve all M1, M2, M3, M4, and M5 behavior.

3. Review and Export Model Decision

M6 separates quality inspection from generated documentation. Review is read-only and safe to run repeatedly in CI or during human inspection. Export is the only command in M6 that creates human-readable markdown artifacts, and those artifacts are generated views, not canonical state.

Review findings describe workflow health. They must distinguish between invalid or inconsistent state and ordinary workflow position. A freshly initialized project, a context-ready project that has not yet been planned, and an in-progress work unit waiting for checkpoint are normal states, not failed reviews.

4. Commands in Scope

```
aiqt review
aiqt review --json

aiqt export project-plan
aiqt export status-report
aiqt export technical-spec
aiqt export agent-packet
aiqt export all
aiqt export <target> --json
aiqt export <target> --format markdown
aiqt export <target> --dry-run
aiqt export <target> --dry-run --json
```

Only markdown export format is supported in M6. --format markdown is accepted explicitly and is also the default. Any other format returns exit code 3.

5. Out of Scope

- automatic fixes from aigt review;
- aigt review --apply;
- replanning or editing the work graph;

- changing work-unit status during review;
- resolving needs_review work units;
- creating checkpoints from review output;
- export formats other than markdown;
- PDF, DOCX, HTML, or JSON document exports;
- default AIQT.md, AGENTS.md, CLAUDE.md, docs/, .milestones/, or .tasks/ generation;
- writing exports outside .aiqt/exports/;
- source-code inspection or automatic validation-command execution;
- Claude/OpenAI/GitHub API integration;
- multi-user collaboration, cloud sync, dashboard, or hosted backend.

6. Behavioral Position in the Workflow

M1: init/status/next -> M2: update -> M3: plan -> M4: next packet -> M5: checkpoint -> M6: review/export

After M5, state may contain completed work, needs_review work, newly ready work, planned downstream work, or an all-done state ready for export. M6 does not decide implementation completion by inspecting source code. It reads canonical state and reports what should happen next.

aiqt review can run at any workflow stage after initialization. It must not require a planned graph or a checkpoint. It returns failures only for invalid state or true blocking review findings, not for normal workflow stages.

7. Review Preconditions

Precondition	Failure behavior
.aiqt/ exists and canonical files are readable	Exit code 3, status failed, recommend aiqt init or repair state.
project.json and state.json validate against schema and version rules	Exit code 3, status failed. No mutation.
runlog.jsonl exists and can be read	Exit code 3 if missing/unreadable. Malformed individual lines become warnings and do not block if project/state are valid.
No workflow-stage requirement beyond initialization	Review proceeds. Missing graph, incomplete context, in-progress work, and no checkpoints are reported as non-blocking workflow-position findings where appropriate.

8. Review Rules

Review rules produce ReviewFinding records. Rules may be blocking or non-blocking. Non-blocking findings can still drive nextRecommendedCommand, but they must not force exit code 1.

8.1 Canonical state integrity findings

Rule	Severity	Blocking
Broken milestoneId on any work unit	critical	true
Dependency references unknown work unit	critical	true
currentWorkUnitId references a missing work unit	critical	true
lastAgentPacket references a missing current or historical work unit	high	true
Checkpoint references a missing work unit or packet ID	high	true

8.2 Workflow-position findings

Rule	Severity	Blocking	Recommended command
workGraph is empty and planningContextReady is false	medium	false	aiqt update
workGraph is empty and planningContextReady is true	medium	false	aiqt plan
currentWorkUnitId is non-null and references an in_progress work unit	medium	false	aiqt checkpoint
At least one work unit has status needs_review	high	false	aiqt review
At least one ready work unit exists and no current work unit is active	info	false	aiqt next
All work units are done	info	false	aiqt export status-report

These are normal workflow states. They help the user navigate the workflow and must not by themselves make aiqt review fail.

8.3 Context and quality findings

Review must not fail a normal early-stage project just because planned work units are not fully enriched. Work-unit quality findings become blocking only when the work unit is actionable or already executed.

Rule	Applies to	Severity	Blocking
Project objective is empty	project	medium	false
No target users are recorded	project	medium	false
No requirements, constraints, technology preferences, business rules, or architecture notes exist	project	medium	false
Open question with status open and impact blocking exists	project	high	true
Accepted requirement has no acceptance criteria	accepted requirement	high	true
Planned work unit has no validationCommands	workUnit.status = planned	medium	false
Planned work unit has no acceptanceCriteria	workUnit.status = planned	medium	false
Planned work unit has vague or empty scope/outOfScope/suggestedFiles	workUnit.status = planned	medium	false
Actionable work unit has no validationCommands	ready, in_progress, needs_review, done	high	true
Actionable work unit has no acceptanceCriteria	ready, in_progress, needs_review, done	high	true
Actionable work unit has empty scope or outOfScope	ready, in_progress, needs_review, done	high	true

Actionable means status is ready, in_progress, needs_review, or done. M4 remains the hard handoff gate for ready work units. M6 review reports planned-work-unit quality gaps as non-blocking guidance only.

8.4 Checkpoint and execution findings

Rule	Severity	Blocking
Work unit done but has no checkpoint record	high	true

Work unit needs_review with open high/critical checkpoint issue	high	false
Last packet exists, currentWorkUnitId is null, and no checkpoint references lastAgentPacket.id	medium	false
Validation result failed or partial on latest checkpoint for a done work unit	high	true
Acceptance criteria result failed or partial on latest checkpoint for a done work unit	high	true

9. Review Finding Model

```
export const ReviewFindingSchema = z.object({
  id: z.string(),
  category: z.enum(["state", "workflow", "context", "graph", "integrity", "checkpoint", "quality", "export"]),
  severity: z.enum(["critical", "high", "medium", "low", "info"]),
  blocking: z.boolean(),
  title: z.string(),
  message: z.string(),
  relatedIds: z.array(z.string()),
  suggestedAction: z.string(),
  nextRecommendedCommand: z.string().nullable(),
});
```

9.1 FIND ID behavior

ReviewFinding.id uses FIND-001, FIND-002, etc. IDs are generated per aiqt review command execution after the deterministic finding sort has been applied. They are ephemeral, are not persisted, and do not continue from previous review runs. M6 must not scan runlog or state for prior FIND IDs.

9.2 Deterministic finding sort order

Review findings must be sorted deterministically before FIND IDs are assigned and before output is rendered.

1. blocking first: true before false;
2. severity order: critical, high, medium, low, info;
3. category lexicographically;
4. relatedIds joined with comma and compared lexicographically;
5. stable rule key or title lexicographically as the final tie-breaker.

After sorting, assign FIND-001, FIND-002, etc. Output order is FIND ID order.

10. Review Command Behavior

1. Locate .aiqt/.
2. Read project.json and state.json.
3. Validate schemas and versions.
4. Inspect runlog health using existing behavior.
5. Evaluate planningContextReady using the M2 helper.

6. Collect review findings from state integrity, workflow position, context, quality, graph, checkpoint, and export-readiness rules.
7. Sort findings deterministically and assign FIND IDs.
8. Compute severity counts and blocking count.
9. Determine nextRecommendedCommand using the single precedence table in Section 17.1.
10. Return human-readable output or CommandResult JSON.
11. Do not mutate project.json, state.json, runlog.jsonl, or exports.

10.1 Review status and exit-code outcomes

Condition	Status	Exit code	Next command
Canonical files missing, malformed, or unsupported schema version	failed	3	aiqt init, aiqt status, or null depending on readable state
One or more blocking review findings	failed	1	aiqt review
No blocking findings and one or more non-blocking findings	warning	0	Use Section 17.1 precedence
No blocking findings and graph is empty but planningContextReady is false	warning	0	aiqt update
No blocking findings and graph is empty but planningContextReady is true	warning	0	aiqt plan
No findings	passed	0	Use Section 17.1 precedence

11. Export Preconditions

Precondition	Failure behavior
.aiqt/ exists and canonical files are readable	Exit code 3, status failed, recommend aiqt init or repair state.
project.json and state.json validate against schema and version rules	Exit code 3, status failed.
Export target is supported	Exit code 3 for unsupported target.
Export format is markdown or omitted	Exit code 3 for unsupported format.
Target-specific source state is available	Exit code 2 for blocked unavailable single target; warning and skip for unavailable optional targets under all.

12. Export Modes and Targets

Target	Filename	Availability
project-plan	.aiqt/exports/project-plan.md	Available after initialization.
status-report	.aiqt/exports/status-report.md	Available after initialization.
technical-spec	.aiqt/exports/technical-spec.md	Available after initialization; content is derived from recorded context, requirements, decisions, graph, and checkpoints.
agent-packet	.aiqt/exports/agent-packet-<packetId>.md	Requires state.lastAgentPacket. Single-target export blocks if unavailable.
all	All available target filenames	Exports project-plan, status-report, technical-spec, and agent-packet if

		available. Unavailable optional targets are skipped with warnings.
--	--	--

12.1 Dry-run mode

--dry-run returns planned export metadata without writing files and without appending runlog events. It must still validate canonical files, target, format, and target availability.

13. Export Command Behavior

This gate order is authoritative for aigt export.

1. Locate .aiqt/.
2. Read project.json and state.json.
3. Validate schemas and versions.
4. Inspect runlog health.
5. Validate export target. Missing target returns exit code 10 with nextRecommendedCommand = aigt export status-report. Unsupported target returns exit code 3.
6. Validate export format. Unsupported format returns exit code 3.
7. Validate target-specific availability. status-report, project-plan, and technical-spec can be generated after initialization; agent-packet requires state.lastAgentPacket; all skips unavailable optional targets with warnings unless no target can be exported.
8. If --dry-run, return planned document metadata without writing files or appending runlog events.
9. Render selected export document or documents.
10. Write only under .aiqt/exports/.
11. Append export.generated runlog event only if files are written.
12. Return deterministic CommandResult.

Invalid target and invalid format are checked before target availability. For example, aigt export made-up --format pdf fails as invalid target before checking whether any export content exists.

14. Export Rendering Rules

- Every generated markdown file must include a generated-file marker at the top.
 - Every generated markdown file must state that canonical state remains .aiqt/project.json and .aiqt/state.json.
 - Exports must not be read back as source of truth by M6 commands.
 - Exports must use deterministic heading order and stable list ordering.
 - Exports must not include full runlog history unless specifically summarized.
 - Exports must not create files outside .aiqt/exports/.
- ```
<!-- Generated by AIQT. Do not edit as source of truth. -->
<!-- Source of truth: .aiqt/project.json and .aiqt/state.json -->
```

### 14.1 Export content minimums

These minimums are authoritative. Implementations may include additional compact sections, but must not omit the listed sections for a supported target.

| Export target  | Required minimum sections                                                                                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| project-plan   | Project Objective; Target Users; Requirements; Constraints; Non-Goals; Open Questions; Milestones; Work Units; Dependencies; Next Recommended Command                                               |
| technical-spec | Technical Overview; Architecture Notes; Technology Preferences; Business Rules; Constraints; Quality Expectations; Work Graph Implementation Plan; Validation Expectations; Suggested Files / Areas |
| status-report  | Project Status; Current Milestone; Current Work Unit; Work Unit Counts; Milestones Table; Work Units Table; Checkpoints Summary; Findings Summary; Next Recommended Command                         |
| agent-packet   | Last Agent Packet Metadata; Work Unit ID; Milestone ID; Content Hash; Created Timestamp; Packet content if available from canonical state                                                           |

## 14.2 Deterministic export ordering

- Milestones render in state.workGraph.milestones stored order.
- Work units render in state.workGraph.workUnits stored order.
- Dependencies render in state.workGraph.dependencies stored order.
- Checkpoints render in state.checkpoints stored order.
- Findings embedded in exports use the same deterministic review finding order as aiqt review.

## 15. Canonical Mutation Rules

| Command                        | Mutates project.json | Mutates state.json | Mutates runlog.jsonl         | Writes exports                 |
|--------------------------------|----------------------|--------------------|------------------------------|--------------------------------|
| aiqt review                    | No                   | No                 | No                           | No                             |
| aiqt export <target> --dry-run | No                   | No                 | No                           | No                             |
| aiqt export <target>           | No                   | No                 | Yes, append export.generated | Yes, under .aiqt/exports/ only |

M6 must not modify workflow status, current pointers, workGraph, checkpoints, lastAgentPacket, or project context. Review is strictly read-only. Export writes generated views only.

## 16. Runlog Rules

aiqt review does not append runlog events. aiqt export appends one export.generated event only when it writes one or more files. Dry-run and failed exports do not append events.

```
{
 "id": "EVT-020",
 "type": "export.generated",
 "timestamp": "ISO_DATE",
 "actor": "aiqt",
 "summary": "Generated AIQT export documents.",
 "relatedIds": ["PROJECT-001"],
 "data": {
 "target": "status-report",
 "format": "markdown",
 "files": [".aiqt/exports/status-report.md"],
 "dryRun": false
 }
}
```

## 16.1 Multi-file export.generated example

When aiqt export all writes more than one file, M6 still appends exactly one export.generated event. The files array contains every written file in deterministic target order. skippedTargets records unavailable optional targets such as agent-packet when state.lastAgentPacket is null.

```
{
 "id": "EVT-021",
 "type": "export.generated",
 "timestamp": "ISO_DATE",
 "actor": "aiqt",
 "summary": "Generated 3 export documents.",
 "relatedIds": ["PROJECT-001"],
 "data": {
 "target": "all",
 "format": "markdown",
 "files": [
 ".aiqt/exports/project-plan.md",
 ".aiqt/exports/status-report.md",
 ".aiqt/exports/technical-spec.md"
],
 "skippedTargets": ["agent-packet"],
 "dryRun": false
 }
}
```

- Event IDs continue from the highest existing EVT id.
- relatedIds includes PROJECT-001 and may include milestone, work unit, checkpoint, or packet IDs when target-specific exports are generated.
- Malformed runlog lines continue to produce warnings under existing runlog health behavior.

## 17. State Outcome and Next Command Rules

### 17.1 Authoritative nextRecommendedCommand precedence for aiqt review

| Priority | Condition                                                             | nextRecommendedCommand                               |
|----------|-----------------------------------------------------------------------|------------------------------------------------------|
| 1        | Canonical state cannot be read or validated                           | aiqt init, aiqt status, or null depending on failure |
| 2        | Any blocking review finding exists                                    | aiqt review                                          |
| 3        | currentWorkUnitId is non-null and references an in_progress work unit | aiqt checkpoint                                      |
| 4        | Any work unit has status needs_review                                 | aiqt review                                          |
| 5        | workGraph is empty and planningContextReady is false                  | aiqt update                                          |
| 6        | workGraph is empty and planningContextReady is true                   | aiqt plan                                            |
| 7        | At least one ready work unit exists and no current work is active     | aiqt next                                            |
| 8        | All work units are done and no blocking review finding exists         | aiqt export status-report                            |
| 9        | No immediate workflow action can be determined                        | aiqt review                                          |

Export is recommended only when the workflow is stable enough to report, when all work is done, or when the user explicitly requested export. It is not recommended as the next fix for needs\_review or invalid-state situations.

### 17.2 Authoritative nextRecommendedCommand rules for aiqt export

| Export outcome | nextRecommendedCommand |
|----------------|------------------------|
|----------------|------------------------|

|                                                                 |                                                                                                                          |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dry-run success                                                 | The current workflow-state recommendation computed from Section 17.1, without treating export itself as the next action. |
| Write success while currentWorkUnitId is non-null               | aiqt checkpoint                                                                                                          |
| Write success with any needs_review work                        | aiqt review                                                                                                              |
| Write success with ready work and no blocking review finding    | aiqt next                                                                                                                |
| Write success with empty graph and planningContextReady = false | aiqt update                                                                                                              |
| Write success with empty graph and planningContextReady = true  | aiqt plan                                                                                                                |
| Write success with all work done                                | aiqt export status-report                                                                                                |
| Target unavailable but soft-skipped under all                   | Same workflow-state precedence as above after warnings are recorded.                                                     |
| Invalid target or invalid format                                | null                                                                                                                     |

Export success must not default to null unless the input is invalid or no workflow-state recommendation can be computed. Export does not change workflow state, so it derives the next recommendation from the current state.

## 17.3 Review output counts

| Field                | Meaning                                                                       |
|----------------------|-------------------------------------------------------------------------------|
| findingCount         | Total number of ReviewFinding records.                                        |
| blockingFindingCount | Number of findings where blocking = true.                                     |
| warningFindingCount  | Number of non-blocking findings with severity critical, high, medium, or low. |
| infoFindingCount     | Number of findings with severity info.                                        |

## 17.4 recommendedExportTargets computation

recommendedExportTargets is computed by aiqt review to help users choose useful generated views. It is advisory and does not mutate state.

| Condition                                  | Targets to include           |
|--------------------------------------------|------------------------------|
| Project has no work graph                  | project-plan                 |
| Work graph exists and no checkpoints exist | project-plan, technical-spec |
| One or more checkpoints exist              | status-report, project-plan  |
| state.lastAgentPacket exists               | agent-packet                 |
| state.projectStatus = review               | status-report                |

Deduplicate recommendedExportTargets in this stable order: project-plan, technical-spec, status-report, agent-packet. M6 does not support a project-summary export target. Do not emit project-summary in human output, JSON output, tests, or export target parsing.

# 18. Data and Schema Contracts

## 18.1 ReviewResultData

```
export interface ReviewResultData {
 findingCount: number;
 blockingFindingCount: number;
 warningFindingCount: number;
 infoFindingCount: number;
 findings: ReviewFinding[];
 runlogHealth: RunlogHealth;
 recommendedExportTargets: string[];
}
```

## 18.2 ExportResultData

```
export interface ExportResultData {
 target: "project-plan" | "status-report" | "technical-spec" | "agent-packet" | "all";
 format: "markdown";
 dryRun: boolean;
 documents: Array<{
 target: string;
 path: string;
 wouldWrite: boolean;
 written: boolean;
 skipped: boolean;
 skipReason: string | null;
 }>;
}
```

## 19. Error Handling and Exit Codes

| Case                                      | Exit code | Status      | Mutation         | Next command                           |
|-------------------------------------------|-----------|-------------|------------------|----------------------------------------|
| Review success with no findings           | 0         | passed      | No               | Section 17.1                           |
| Review with non-blocking findings only    | 0         | warning     | No               | Section 17.1                           |
| Review with blocking findings             | 1         | failed      | No               | aiqt review                            |
| Missing .aiqt/ or invalid canonical state | 3         | failed      | No               | aiqt init, aiqt status, or null        |
| Export missing target                     | 10        | needs_input | No               | aiqt export status-report              |
| Export unsupported target or format       | 3         | failed      | No               | null                                   |
| Export target unavailable, single target  | 2         | blocked     | No               | aiqt review or aiqt next as applicable |
| Export dry-run success                    | 0         | passed      | No               | Section 17.2                           |
| Export write success                      | 0         | passed      | Exports + runlog | Section 17.2                           |

## 20. Repository Implementation Scope

```
src/
 cli/
 commands/
 review.command.ts
 export.command.ts
 services/
 review-service.ts
 export-service.ts
 workflow/
 review-rules.ts
 review-findings.ts
 review-next-command.ts
 templates/
 exports/
 project-plan.template.ts
 status-report.template.ts
 technical-spec.template.ts
 agent-packet-export.template.ts
```

```

state/
 runlog-store.ts # extend with export.generated builder if needed
core/
 filesystem/
 safe-writer.ts # reuse for export writes where practical

tests/
 unit/
 review-service.test.ts
 review-findings.test.ts
 review-next-command.test.ts
 export-service.test.ts
 export-templates.test.ts
 integration/
 review.command.test.ts
 export.command.test.ts
 review-export-flow.test.ts
 fixtures/
 review/
 export/

```

Follow the actual repository layout produced by M1-M5. Do not create duplicate helper locations if equivalent folders already exist.

## 21. Runtime and Package Impact

- Continue using Node.js, TypeScript, ESM, pnpm, commander, zod, and vitest.
- Do not add markdown-parser, PDF, DOCX, HTML rendering, AI, GitHub, or source-code execution dependencies in M6.
- Use simple string templates for markdown exports.
- package.json.version may remain unchanged unless implementation intentionally bumps package metadata.
- AIQT\_SCHEMA\_VERSION should remain 0.5.0 unless final implementation review identifies an incompatible canonical schema change.

## 22. Testing Scope

- All existing M1-M5 tests must continue to pass.
- Review unit tests must cover state integrity findings, workflow-position findings, context/quality findings, checkpoint findings, finding sorting, FIND ID generation, and next-command precedence.
- Review integration tests must cover initialized-only, context-ready-no-graph, planned-ready-work, in-progress-current-work, needs\_review, all-done, invalid graph references, malformed runlog line warnings, and JSON output shape.
- Review regression tests must prove planned work units with missing validationCommands or acceptanceCriteria produce non-blocking findings, while ready work units with the same gaps produce blocking findings.
- Export unit tests must cover target validation, format validation, availability rules, export nextRecommendedCommand rules, content minimums, dry-run metadata, generated-file markers, deterministic rendering, and all target skip warnings.
- Export integration tests must cover each target, --dry-run, --json, unsupported target, unsupported format, agent-packet unavailable, all with packet unavailable, and no writes outside .aiqt/exports/.
- Tests must confirm aiqt review never mutates files.
- Tests must confirm export writes only under .aiqt/exports/ and appends runlog only on actual writes.

## 23. Implementation Tasks

| Task   | Description                                                                                             |
|--------|---------------------------------------------------------------------------------------------------------|
| M6-T1  | Add review command registration, options, and CommandResult wiring.                                     |
| M6-T2  | Implement ReviewFinding schema/model, deterministic sorting, and per-run FIND ID assignment.            |
| M6-T3  | Implement review rule collectors for state, workflow, context, graph, checkpoint, and export readiness. |
| M6-T4  | Implement authoritative nextRecommendedCommand precedence for review.                                   |
| M6-T5  | Implement review human and JSON output.                                                                 |
| M6-T6  | Add export command registration, target/format parsing, and gate order.                                 |
| M6-T7  | Implement export renderers and generated-file markers.                                                  |
| M6-T8  | Implement dry-run behavior and export write behavior under .aiqt/exports/.                              |
| M6-T9  | Append export.generated runlog events only on write success.                                            |
| M6-T10 | Add full unit and integration tests and preserve M1-M5 test suite.                                      |

## 24. Definition of Done

- aiqt review and aiqt review --json work against any initialized project.
- Normal workflow states are reported as non-blocking findings and do not fail review.
- Blocking review findings produce status failed and exit code 1.
- Invalid canonical state produces exit code 3.
- Review findings are sorted deterministically and use per-run FIND IDs.
- Review nextRecommendedCommand is computed from the single precedence table.
- Planned-work-unit quality gaps are non-blocking; actionable-work-unit quality gaps are blocking.
- recommendedExportTargets is computed deterministically using only valid M6 export targets.
- aiqt export supports project-plan, status-report, technical-spec, agent-packet, and all.
- Export target/format/availability gates and export nextRecommendedCommand rules are deterministic.
- Each export target satisfies the M6 v0.4 content minimum table.
- Export --dry-run writes nothing and appends no runlog event.
- Export writes only under .aiqt/exports/ and includes generated-file markers.
- No AIQT.md, AGENTS.md, CLAUDE.md, docs/, .milestones/, or .tasks/ files are created by default.
- pnpm typecheck, pnpm lint, pnpm test, pnpm build, and pnpm validate all pass.

## 25. Agent Execution Packet

# AGENT EXECUTION PACKET

## Role

You are implementing Milestone 6 for AIQT CLI.

## Starting Point

M5 is complete and merged to main.

Start from: main

Current main commit: 41a8e49

Required existing tag: m5-checkpoint-engine

Create branch: feat/m6-review-export-engine

## Current Work Unit

Milestone: M6 - Review and Export Engine

Objective: Implement aiqt review and aiqt export as the final MVP workflow loop commands.

#### ## Scope

- Implement aiqt review and aiqt review --json.
- Implement aiqt export targets and --dry-run.
- Keep review read-only and CI-safe.
- Write exports only under .aiqt/exports/.
- Add deterministic review findings, FIND IDs, sort order, and next-command precedence.
- Add tests covering the M6 v0.4 matrix.

#### ## Out of Scope

- Automatic fixes, replanning, source inspection, agent execution, external APIs, and non-markdown exports.

#### ## Validation Commands

```
```bash
pnpm typecheck
pnpm lint
pnpm test
pnpm build
pnpm validate
```
```

#### ## Required Agent Output

Return:

1. Branch name
2. Commit hash
3. Summary of implementation
4. Files created
5. Files modified
6. Tests added or changed
7. Validation commands run and results
8. Confirmation that M1-M5 tests still pass
9. Confirmation that no out-of-scope functionality was implemented
10. Any unresolved issues
11. Suggested next step

## Appendix A - Minimum Test Matrix

| Area   | Scenario                                                                | Expected result                                                                     |
|--------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Review | Initialized project immediately after aiqt init                         | warning, exit 0, non-blocking workflow finding, recommend aiqt update               |
| Review | Context ready but graph empty                                           | warning, exit 0, non-blocking workflow finding, recommend aiqt plan                 |
| Review | Planned work unit missing validationCommands                            | warning, exit 0, non-blocking work-unit quality finding with category = quality     |
| Review | Ready work unit missing validationCommands                              | failed, exit 1, blocking work-unit quality finding with category = quality          |
| Review | currentWorkUnitId references in _progress work unit                     | warning, exit 0, non-blocking workflow finding, recommend aiqt checkpoint           |
| Review | needs_review work exists and another ready work unit exists             | warning or failed only if blocking findings exist; recommend aiqt review            |
| Review | broken dependency reference discovered during review integrity analysis | failed, exit 1, critical blocking integrity finding; no mutation                    |
| Review | malformed project.json                                                  | failed, exit 3                                                                      |
| Review | malformed runlog line only                                              | warning, exit 0, runlogHealth exposed                                               |
| Review | finding sort stability                                                  | blocking, severity, category, relatedIds, rule key; FIND IDs assigned after sorting |
| Export | missing target                                                          | needs_input, exit 10                                                                |

|            |                                           |                                                                                |
|------------|-------------------------------------------|--------------------------------------------------------------------------------|
| Export     | unsupported target                        | failed, exit 3                                                                 |
| Export     | unsupported format                        | failed, exit 3                                                                 |
| Export     | status-report dry-run                     | passed, exit 0, no files written, no runlog event                              |
| Export     | status-report write                       | passed, exit 0, file under .aiqt/exports/, export.generated event              |
| Export     | agent-packet unavailable as single target | blocked, exit 2, no mutation                                                   |
| Export     | all with agent-packet unavailable         | passed or warning, generated available docs, skipped agent-packet with warning |
| Regression | M1-M5 validation suite                    | all tests still pass                                                           |

## Appendix B - Example Review JSON Output

```
{
 "status": "warning",
 "action": "review",
 "projectStatus": "draft",
 "currentMilestoneId": null,
 "currentWorkUnitId": null,
 "summary": "Review completed with non-blocking workflow findings.",
 "completedActions": ["Read project.json", "Read state.json", "Evaluated review rules"],
 "changedFiles": [],
 "affectedItems": ["PROJECT-001"],
 "blockingIssues": [],
 "warnings": [],
 "requiresHumanInput": false,
 "nextRecommendedCommand": "aiqt update",
 "data": {
 "findingCount": 2,
 "blockingFindingCount": 0,
 "warningFindingCount": 2,
 "infoFindingCount": 0,
 "recommendedExportTargets": ["project-plan"],
 "findings": [
 {
 "id": "FIND-001",
 "category": "workflow",
 "severity": "medium",
 "blocking": false,
 "title": "Project context is incomplete",
 "message": "The project is initialized but not ready for planning.",
 "relatedIds": ["PROJECT-001"],
 "suggestedAction": "Run aiqt update to capture project context.",
 "nextRecommendedCommand": "aiqt update"
 },
 {
 "id": "FIND-002",
 "category": "context",
 "severity": "medium",
 "blocking": false,
 "title": "No target users recorded",
 "message": "Planning quality improves when at least one target user is recorded.",
 "relatedIds": ["PROJECT-001"],
 "suggestedAction": "Add target users through aiqt update.",
 "nextRecommendedCommand": "aiqt update"
 }
]
 }
},
```

```
"exitCode": 0
}
```

## Appendix C - Export Markdown Templates

### C.1 Common generated-file header

```
<!-- Generated by AIQT. Do not edit as source of truth. -->
<!-- Source of truth: .aiqt/project.json and .aiqt/state.json -->
```

### C.2 status-report.md

```
AIQT Status Report

Project
- Name: <project.name>
- Objective: <project.objective>
- Status: <state.projectStatus>

Current Workflow Position
- Current milestone: <state.currentMilestoneId or None>
- Current work unit: <state.currentWorkUnitId or None>
- Next recommended command: <state.nextRecommendedCommand>

Work Unit Counts
- <status>: <count>

Milestones Table
| ID | Title | Status | Work Units |

Work Units Table
| ID | Milestone | Title | Status | Dependencies |

Checkpoints Summary
- Checkpoints: <count>

Findings Summary
- Blocking findings: <count>
- Non-blocking findings: <count>
```

### C.3 project-plan.md

```
AIQT Project Plan

Project Objective
<project.objective>

Target Users
<target users>

Requirements
<requirements>

Constraints
<constraints>

Non-Goals
<non-goals>

Open Questions
```

<open questions>

## Milestones  
<ordered milestones>

## Work Units  
<ordered work units>

## Dependencies  
<dependencies>

## Next Recommended Command  
<nextRecommendedCommand>

## C.4 technical-spec.md

# AIQT Technical Specification Export

## Technical Overview  
<technical overview>

## Architecture Notes  
<architecture notes>

## Technology Preferences  
<technology preferences>

## Business Rules  
<business rules>

## Constraints  
<constraints>

## Quality Expectations  
<quality expectations>

## Work Graph Implementation Plan  
<work graph plan>

## Validation Expectations  
<validation commands and expectations>

## Suggested Files / Areas  
<suggested files or areas>

## C.5 agent-packet export

# AIQT Agent Packet Export

Packet ID: <lastAgentPacket.id>  
Work Unit ID: <lastAgentPacket.workUnitId>  
Created At: <lastAgentPacket.createdAt>  
Content Hash: <lastAgentPacket.contentHash>

This export records packet metadata only unless the implementation stores packet text in state. M6 must not invent packet text that is not available from canonical state.